RAJALAKSHMI ENGINEERING COLLEGE
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Register No.: 251801057

Continuous Assessment-II		
Artificial Intelligence and Machine Learning (III sem)		
Degree: B. E / B. Tech	Semester: III &V	Year: II &III
Course code & Title: AI23231 & Principles of Artificial Intelligence		
Date: 28.10.26		QP CODE: 151102
[Common to: AIDS (III sem) & IT (V sem)]		

[Regulations: 2023]

Time: 2 hr 15min Max. Marks: 75

Answer All Questions

Part A: 10 x 2 = 20 Marks

1	Relate the concept of an optimal strategy to the utility obtained by a player in a competitive game.	[CO3]	[K2]
2	A MIN node receives the backed-up values 6, 4 and 8 from its children. Find the value propagated to its parent.	[CO3]	[K3]
3	What is meant by knowledge representation in Artificial Intelligence?	[CO4]	[K1]
4	Compare a fact and a rule in a rule-based knowledge representation system.	[CO4]	[K2]
5	Define procedural knowledge and state one situation in which it is useful.	[CO4]	[K1]
6	Relate a goal in logic programming to the inference process used to establish whether the goal is true.	[CO4]	[K2]
7	Define conditional probability and state its role in probabilistic reasoning.	[CO5]	[K1]
8	Compare a Bayesian Network and a Decision Network with respect to the type of information represented.	[CO5]	[K2]
9	Relate belief and plausibility in Dempster–Shafer theory to the representation of uncertain evidence.	[CO5]	[K2]
10	An action has outcomes with utilities 10 and 4, occurring with probabilities 0.6 and 0.4. Find its expected utility.	[CO5]	[K3]

Part B: 5x11=55 Marks

11 a. Discuss the minimax search procedure used in adversarial game playing.Explain the roles of MAX and MIN players and describe how utility values are propagated through a game tree. [CO3] [K3]

[OR]

b. Discuss the concept and working of alpha-beta pruning in game-tree search.Explain how alpha and beta values are updated during traversal and how unnecessary branches are eliminated. [CO3] [K3]

12 a. An intelligent library assistant must represent the following knowledge: [CO4] [K3]

- Every research scholar is a postgraduate student.
- Postgraduate students can borrow research books.
- Faculty members can borrow reference books.
- A student with overdue books cannot borrow additional books.
- Meena is a research scholar.
- Meena has no overdue books.

The knowledge relationships are shown below.

Research Scholar

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v

Postgraduate Student

|

v

Borrow Research Books

Faculty Member ------> Borrow Reference Books

Overdue Books ------X------> Additional Borrowing

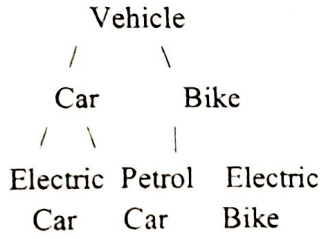
(i) Convert the given statements into a suitable set of facts and rules for a rule-based knowledge representation system. (4 marks)

(ii) Determine what borrowing privileges can be inferred for Meena and justify each conclusion from the rule base. (3 marks)

(iii) Analyse whether the above knowledge is better represented purely as declarative knowledge or whether some procedural knowledge is also required in a practical library system. Support your answer with suitable examples. (4 marks)

[OR]

b. Consider the following knowledge hierarchy: [CO4] [K3]



Assume the following knowledge:

- Every Vehicle has a registration number.
- Car has four wheels.
- Bike has two wheels.
- Electric vehicles use batteries.
- Petrol Car uses fuel.
- Electric Car has regenerative braking.

(i) Construct a structured knowledge representation showing the classes, subclasses, attributes and inherited properties. (4 marks)

(ii) Determine the complete set of inherited and explicitly defined properties of an Electric Car and an Electric Bike. (4 marks)

(iii) Analyse how inheritance avoids duplication of knowledge in this representation and identify one case where property overriding may be required. (3 marks)

13 a. A diagnostic expert system contains the following facts and rules. [CO4] [K2]

Initial Facts

- Fever(Ravi)
- Cough(Ravi)
- HighWBC(Ravi)

Rules

- R1: Fever(x) ∧ Cough(x) → RespiratoryInfection(x)
- R2: RespiratoryInfection(x) ∧ HighWBC(x) → BacterialInfection(x)
- R3: BacterialInfection(x) → AntibioticTreatment(x)
- R4: Fever(x) ∧ Rash(x) → ViralInfection(x)
- R5: ViralInfection(x) → AntiviralTreatment(x)

Initial Facts

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v

Rule R1

|

v

Respiratory Infection

|

v

Rule R2

|

v

Bacterial Infection

|

v

Rule R3

|

v

Treatment

(i) Apply forward reasoning to the given facts and rules. Show the contents of the working memory after every successful rule firing. (5 marks)

(ii) Identify the rules that do not fire and state why their antecedent conditions are not satisfied. (3 marks)

(iii) Determine the final treatment recommendation and examine whether it can still be derived if the fact HighWBC(Ravi) is removed. (3 marks)

[OR]

b. Using the same domain, suppose the system has the goal: [CO4] [K2]

AntibioticTreatment(Ravi)

The knowledge base contains:

- R1: Fever(x) ∧ Cough(x) → RespiratoryInfection(x)
- R2: RespiratoryInfection(x) ∧ HighWBC(x) → BacterialInfection(x)
- R3: BacterialInfection(x) → AntibioticTreatment(x)

Known facts:

- Fever(Ravi)
- Cough(Ravi)
- HighWBC(Ravi)

Goal

AntibioticTreatment(Ravi)

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v

BacterialInfection(Ravi)

|

v

RespiratoryInfection(Ravi)

/ \

v v

Fever & Cough HighWBC

(i) Apply backward reasoning beginning with the goal *AntibioticTreatment(Ravi)* and construct the complete sequence of sub-goals generated. (5 marks)

(ii) Match each sub-goal with the appropriate rule and fact until the original goal is proved. (3 marks)

(iii) Determine whether the goal can be proved if Cough(Ravi) is absent from the fact base, and justify the result. (3 marks)

14 a. Discuss the structure and working of a Bayesian Network. Explain how nodes, directed edges, conditional dependencies, and conditional probability tables are used to represent uncertain knowledge. [CO5] [K2]

[OR]

b. Explain the basic concepts of Dempster–Shafer Theory. Discuss belief, plausibility, mass functions, and the combination of evidence from multiple sources. [CO5] [K2]

15 a. Discuss the concept of a Markov Decision Process (MDP). Explain its components—states, actions, transition probabilities, rewards, and policy—and describe how an agent selects actions to maximize long-term reward. [CO5] [K4]

[OR]

b. Discuss the concept of fuzzy logic and explain how it differs from classical logic. Describe fuzzification, membership functions, rule evaluation, and defuzzification with a suitable example. [CO5] [K4]